

Handout 3: Elasticity and Its Application

Mankiw Ch 5

Problem 1. Your websites: how many fewer, and what happens to revenue

You build websites for local businesses. At \$200 a site you sell 12 a month. You are thinking about raising the price to \$250, and you expect to sell 8 a month at that price.

- a) Compute the price elasticity of demand by dividing by the starting values: first for the move from \$200 to \$250, then for the move from \$250 back to \$200. Why do the two answers differ?

- b) Compute the elasticity by the midpoint method. Is your demand elastic or inelastic?

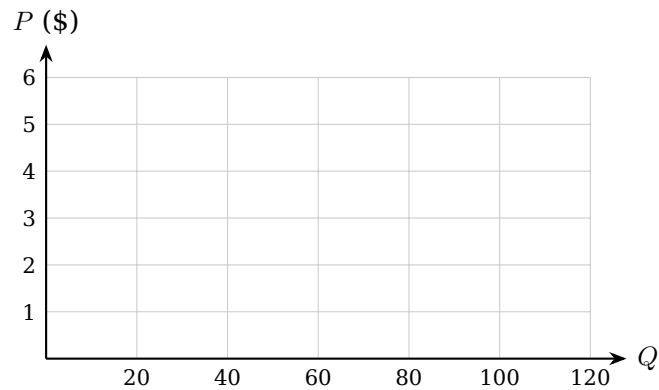
- c) Compute your total revenue at each price. Does the price increase raise revenue? Is that consistent with your answer to (b)?

- d) Suppose instead that you would still sell 10 sites a month at \$250. Recompute the midpoint elasticity and the revenue. What changed?

Problem 2. Revenue along a straight-line demand curve

A taco truck near campus faces daily demand $Q_d = 120 - 20P$, where P is the price of a taco in dollars.

P (\$)	Q_d	$TR = P \times Q_d$ (\$)
1		
2		
3		
4		
5		



- Fill in the table and draw the demand curve on the grid. At what price is total revenue largest?
- Compute the midpoint elasticity between \$1 and \$2, and between \$4 and \$5.
- The truck charges \$4 and wants more revenue. Should it raise or lower its price? What if it charges \$2?
- Every step down this curve is the same: \$1 off the price, 20 more tacos. Why does the elasticity change along it?

Problem 3. Income, cross-price, and supply elasticities

For (a)–(e), compute the elasticity, keeping its sign, and classify the good or the pair of goods.

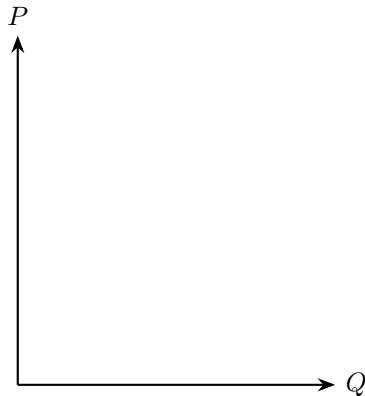
- a) Incomes rise 5 percent; purchases of store-brand cereal fall 2 percent.
- b) Incomes rise 5 percent; the number of international flights taken rises 12 percent.
- c) Incomes rise 5 percent; household electricity use rises 1 percent.
- d) The price of Uber rides rises 15 percent; the number of Lyft rides rises 9 percent.
- e) The price of game consoles falls 10 percent; sales of video games rise 5 percent.
- f) The price of fresh strawberries rises from \$3 to \$5 a pint, and within the week growers bring 1,100 pints to market instead of 900. Compute the price elasticity of supply by the midpoint method. Would you expect it to be larger or smaller if the price stayed at \$5 for three years? Why?

Problem 4. Two ways to fight drug use

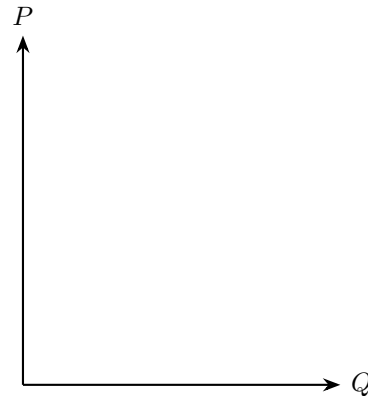
Two policies are proposed to reduce the use of illegal drugs:

- (1) Increase the number of federal agents devoted to drug interdiction.
- (2) Open more addiction treatment centers and spend more on education.

The demand for illegal drugs is highly inelastic. Use a diagram for each policy.



(1) Interdiction



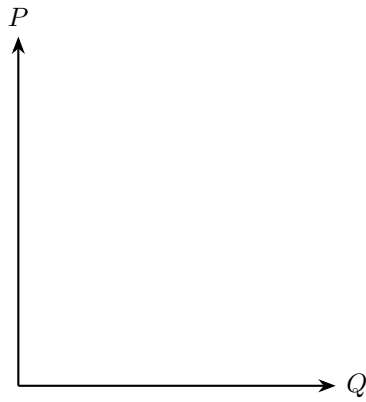
(2) Treatment and education

- a) Policy (1), interdiction. Which curve shifts, and which way? What happens to the price, the quantity, and total spending on drugs?
- b) Policy (2), treatment and education. Same questions.
- c) Which policy reduces the total amount of money spent on drugs? Which is more likely to reduce drug-related crime?
- d) Would your answer about interdiction change over a longer horizon?

Problem 5. A tax to reduce smoking

Studies indicate that the price elasticity of demand for cigarettes is about 0.2. A pack currently costs \$5, and the government wants to reduce smoking.

- a) What price would reduce smoking by 10 percent?
- b) A tax raises the price of cigarettes. Is the policy more effective in the short run or in the long run? Show it on a supply-and-demand graph.



- c) Do teenagers have a higher price elasticity of demand for cigarettes than adults? Discuss using each statement.
- (i) It is legal for adults to drink alcohol, so many adults spend their money on alcohol rather than cigarettes.
 - (ii) Adults are more likely to be addicted to cigarettes.
 - (iii) Teenagers have less income than adults.